FIONA JUSTINAMALADIRAVIAM PATRICIA JOSELYN

VLSI graduate student with 3 years of strong industry experience in ASIC/FPGA RTL Design.

fjustina@purdue.edu |765 746 9680| Locations: Across USA| Languages: English, Tamil| Linkedin

PROFESSIONAL EXPERIENCE

ASIC Design Engineer **Aug 2022 – May 2025**

MBIT Wireless Pvt Ltd, Chennai, India

- Engineered and optimized signal processing and cell search algorithms and FFT in **Verilog** for a custom 4G modem IC achieving low power and low area implementation by using CG, DFS etc in **Synopsys tools and Cadence Genus** for analysis in **65nm**.
- Derived and validated ASIC/FPGA **timing constraints** from scratch for both full chip and 3 subsystems for 2 projects through in-depth architectural analysis in Xilinx and cadence genus by using TCL.
- Proposed architectural modifications and implemented targeted slack fixes via refined constraint, achieving **timing closure** improvements to 50-100ps. Close work with BE teams from RTL to GDS**.**
- Developed 50+ design-oriented new verification/validation scenarios as a designer and developed 10+ standalone TBs from scratch for 100% robust verification/FPGA validation with **functional coverage** of 92%.
- Analysed **netlists for RTL to standard cell mapping** to fix slack in dedicated paths. Analysed **structural and functional CDC**.
- Analysed **synthesis, preplace, place, preCTS and Post CTS timing reports from Innovus** for timing exceptions, IO modelling and system modelling constraints achieving timing closure improvements to 100ps

SKILLS

Languages: Verilog HDL, TCL, Python, C++, CUDA
Skills: Low power/Area design, Static timing analysis, CDC, Microarchitecture, Scenario listing, Lint, Linux, RISC, Cache, Memory design.
Tools: Synopsys VCS, Xilinx Vivado, Cadence- Genus, Innovus, Virtuoso, Conformal CDC, Github

PROJECTS

Goldschmidt algorithm-based IEEE 32-bit floating point divider

- Implemented the divider algorithm entirely from scratch in Verilog using EDA tools from research papers.
- Designed 9 different arithmetic units from scratch and replaced in the divider implementation.
- Performance analysis using Cadence Genus and observed ~7% decrease in delay than the preexisting design.

Wireless power transmission to Micro UAVs using FPGA control algorithms

- SPI, I2C and UART based communication control using system Verilog on Spartan 7 Xilinx FPGA for programming on board PLL chip for RF beam forming.

Write instruction buffer and victim cache implementation in Gem5 (coursework project)

- Compared the implementation against various benchmark algorithms and achieved 30% speedup in the execution.

8-bit Wallace Tree Multiplier design in 45nm using Cadence virtuoso.

- Developed a Wallace tree circuit design mode using gpdk45 in cadence with 1.5ns worst case delay.
- Compact design with 1000 um area.Robust circuit with 100% functional verification with all corner TCs covered

ACHEIVEMENTS

1. **Gold medallist** B.E-ECE (In a class of 180) — Sri Venkateswara College of Engineering
   Awarded for academic excellence
2. **Best Outgoing ECE student award** — Sri Venkateswara College of Engineering
   Recognition for the best holistic student in the department
3. **Apple's circuit design competition winner** — Purdue university
   Ranked 3rd for designing WTM circuit in virtuoso
   with best PPA and report of metrics.

EDUCATION

Bachelors in Electronics and Communication Engineering **Aug 2018 - May 2022**

Sri Venkateswara College of Engineering, Chennai, India
Specialization: VLSI
Coursework: Digital system design, DSP, VLSI Design and computer organization and design.

Master of Science in electrical and computer engineering **Aug 2025 - May 2027**

Purdue university, West Lafayette, Indiana, USA

Specialization: VLSI and Circuit design
Coursework: MOS VLSI, Computer architecture, Parallel Programmable accelerators.